Education and Student Outcomes
Grade Range from Student Performance/Interaction Logs

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Abstract

Predicting student academic performance is an important problem in education, particularly as many institutions increasingly rely on online and digital learning platforms. These platforms generate large amounts of data describing how students interact with course materials and how their academic progress evolves over time. Because of this, there is a growing interest in applying machine learning techniques to educational data in order to better understand student behavior and outcomes.

In this project we aim to predict whether a student will pass or fail a course using the UCI Student Performance dataset. We compare several baseline machine learning models including logistic regression, support vector machines, and Naive Bayes with a neural network model designed to capture patterns in student data. We present initial experimental results and analyze how well these models identify students who may be at risk of failing. Our goal is to better understand which student behaviors and characteristics are most predictive of academic success and how predictive models could potentially be used responsibly in educational environments.

1 Introduction and Motivation

Predicting whether a student will pass or fail a course is an important challenge in educational data mining. Early identification of students who may struggle academically allows instructors and institutions to provide additional support before the situation becomes critical. Instead of reacting after students have already fallen behind, predictive models allow educators to intervene earlier and help students stay on track.

Educational institutions collect large amounts of information about students, including demographic background, family characteristics, study habits, attendance records, and academic performance throughout the semester. This type of information can potentially provide useful signals about how students are performing in their classes. Machine learning models can analyze these factors to discover patterns that may not be obvious to instructors who are only observing the class from a limited perspective.

Another reason this problem is interesting is that many schools today use online learning systems or digital tools which naturally generate large datasets. These systems track interactions such as assignment submissions, grades, attendance, and sometimes engagement with course materials. While not every dataset contains all of these signals, even basic demographic and academic information can still reveal patterns when analyzed using machine learning methods.

From a machine learning perspective, this task is also interesting because it involves structured tabular data with a mixture of demographic, behavioral, and academic attributes. Similar predictive approaches are widely used in many other domains such as healthcare, finance, and business analytics, where early prediction of risk or performance can lead to improved outcomes.

Our project focuses on identifying which factors contribute most strongly to predicting student success or failure. By applying several machine learning models to a commonly used education dataset, we analyze how effectively these models can capture patterns associated with academic performance. We also investigate how different modeling choices influence the ability to detect students who may be at risk of failing. Understanding these patterns can also help provide insights into how predictive models might be used to support students rather than penalize them.

2 Related Work

Prior work in educational data mining has examined student performance prediction from two primary perspectives: structured demographic features and behavioral engagement data. (Cortez and Silva, 2008) demonstrated that traditional supervised learning models can achieve strong predictive accuracy using tabular features from the UCI Student Performance dataset.

The work by Cortez and Silva focused on predicting student grades using a variety of attributes such as family background, study time, absences, and previous academic performance. Their results showed that even relatively simple machine learning techniques could achieve strong predictive accuracy when applied to structured student data.

More recent work using datasets such as the Open University Learning Analytics Dataset (OULAD) has shown that incorporating engagement patterns from virtual learning environments allows for earlier and more accurate identification of at risk students (Kuzilek et al., 2017). These datasets include information about how frequently students interact with course materials or participate in online learning systems.

Deep learning approaches extend this idea by modeling temporal learning behavior and capturing patterns that simpler models may miss. For example, sequential models such as recurrent neural networks can model how student activity changes over time throughout a course. This can potentially allow models to detect warning signs earlier in the semester.

While these studies demonstrate the value of both structured and behavioral predictors, they often analyze them independently. Our project builds on this prior work by comparing baseline models with neural approaches in order to better understand how model complexity and feature representation influence predictive performance.

3 Dataset

Source and Size

We use the Student Performance dataset from the UCI Machine Learning Repository. It contains two datasets from two Portuguese schools. The first is `student-mat.csv` (Mathematics – 395 students) and the second is `student-por.csv` (Portuguese language – 649 students). Each record represents one student with demographic, family, social, and school-related attributes along with grade outcomes.

Overall this provides slightly over one thousand student records when both datasets are combined. While this is not a massive dataset compared to some modern machine learning benchmarks, it is still large enough to evaluate different classification approaches and observe general trends in student performance.

Type of Data

This dataset is tabular data with a mix of categorical variables such as school, sex, and internet access. It also contains numerical variables including age, study time, and number of absences.

Because the dataset contains both categorical and numeric information, some preprocessing steps are required before training machine learning models. Many algorithms cannot directly interpret categorical values such as text labels, so these must be converted into numerical representations.

Labels and Prediction Target

Our target variable is a binary pass/fail outcome. The dataset provides a final grade variable called `G3` which ranges from 0 to 20. Following standard conventions, a student is considered to have passed if the grade is greater than or equal to 10 and failed if it is less than 10.

Transforming the problem into a binary classification task simplifies the modeling process and allows us to evaluate how well models can distinguish between successful and struggling students.

Preprocessing

We convert the final grade variable into a binary pass/fail label. Yes/no fields are converted into binary values (0 or 1). Categorical variables are transformed using one hot encoding so they can be used by machine learning models. Numerical variables are normalized to ensure that features with larger numeric ranges do not dominate the learning process.

These preprocessing steps are fairly common when working with tabular datasets and help ensure that models treat all features in a more balanced way during training.

Ethical Considerations

This dataset captures student background information and academic factors related to performance. While this makes it useful for predictive modeling research, predictions about student success can be sensitive. Such predictions could potentially reflect social or demographic biases present in the data.

In this project our goal is not to make high stakes decisions about students but rather to explore how predictive models can identify students who may benefit from additional academic support. Any real world system using these types of predictions would need to carefully consider fairness, transparency, and how predictions are communicated to instructors and students.

4 Methodology

We model student performance prediction as a binary classification task, where the objective is to predict whether a student will pass or fail a course based on demographic and academic features.

We first implement several baseline supervised learning models, including Logistic Regression, Support Vector Machines (SVM), and Naive Bayes. These models are commonly used for classification tasks and are well suited for tabular datasets. Logistic regression provides a strong interpretable baseline, while SVM models can capture more complex decision boundaries between classes. Naive Bayes offers a probabilistic approach that assumes independence between features.

Baseline models are important in machine learning experiments because they provide a reference point for evaluating more complex approaches. If a more advanced model does not significantly outperform a simpler baseline, then the additional complexity may not be justified.

In addition to these baseline approaches, we experiment with a neural network model designed to capture more complex patterns in the dataset. Neural networks can model nonlinear relationships between features and may be able to detect subtle patterns that traditional models cannot easily identify.

All models are trained using the same preprocessed dataset to ensure fair comparisons. Categorical variables are one-hot encoded, yes/no variables are converted into binary values, and numerical variables are normalized. Hyperparameters for each model are tuned using the validation set.

5 Evaluation Plan

We evaluate model performance using several standard binary classification metrics including accuracy, precision, recall, and F1-score. Accuracy measures the overall proportion of correct predictions. Precision measures how many predicted positive cases are correct, while recall measures how well the model identifies actual positive cases. The F1-score provides a balanced measure that considers both precision and recall.

These metrics are particularly important in this task because simply maximizing accuracy may not always produce the most useful model. For example, if most students pass the course, a model could achieve high accuracy by predicting that every student will pass.

For our experiments we use an 80/10/10 data split. Eighty percent of the dataset is used for training the models, ten percent is used for validation during hyperparameter tuning, and the remaining ten percent is used as the final test dataset.

This evaluation strategy allows us to measure how well the models generalize to unseen data. We also compare the performance of different logistic regression variations in order to better understand how training strategies influence prediction accuracy.

6 Initial Experimental Results

We used an 80/10/10 split for the experimental results with 80% of the data used to train the models, 10% used for validation, and the remaining 10% used for testing.

We experimented with three different types of logistic regression models for our experiments. The first logistic regression model used a standard dataset split and evaluated predictions on the held out test data. This model performed significantly better when predicting students who would pass compared to students who would fail.

The model correctly predicted students who passed 86.58% of the time, correctly identifying 71 out of the 82 students who passed. However, when predicting failures the model performed much worse. Out of the 23 students who failed, the model correctly predicted only 5 failures, resulting in a failure recall of only 21.7%. The test set confusion matrix in Figure 1 clearly illustrates this asymmetry.

To improve the prediction of failing students we implemented an out-of-fold logistic regression approach using cross validation. This approach attempts to reduce overfitting and improve generalization. With this method the failure recall increased to 52.17%, although the recall for passing students decreased to 73.17%.

In a third experiment we adjusted the classification threshold used for logistic regression predictions. By increasing the threshold to 0.84 we were able to further improve the failure recall to 65.22%. However this change reduced the recall for passing students to 59.76%.

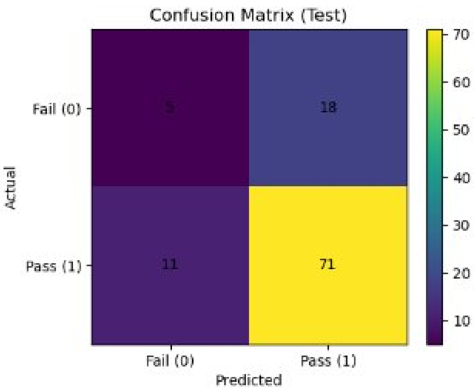

Figure 1: Confusion matrix on the test set for the standard logistic regression model. The model correctly identifies 71 of 82 passing students but only 5 of 23 failing students.

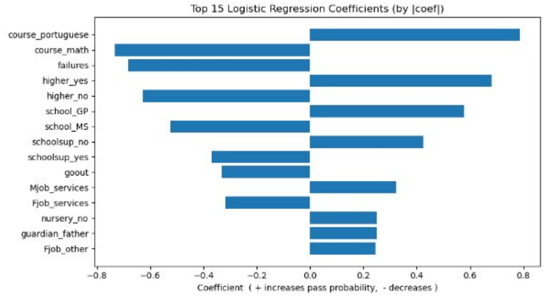

Figure 2: Top 15 logistic regression coefficients ranked by absolute magnitude. Positive values (right) increase predicted pass probability; negative values (left) decrease it.

The F1 scores provide a clearer comparison between the models. The first logistic regression model achieved an F1 score of 83.04%. The out-of-fold logistic regression model achieved an F1 score of 78.43%. The threshold adjusted model achieved an F1 score of 70.5%.

Although the first model struggled to predict failing students, it was extremely accurate when predicting students who would pass. As a result it produced the highest overall F1 score among the tested models.

7 Discussion and Key Observations

One of the most important observations from our experiments is that the models consistently predict passing students more accurately than failing students. This imbalance occurs partly because the dataset itself contains more students who pass than students who fail.

Because of this imbalance, models that optimize for overall accuracy naturally favor predicting the majority class. As a result, models may achieve high accuracy while still performing poorly at identifying failing students.

We also observed that certain features appeared to correlate more strongly with failure outcomes. Figure 2 shows the top 15 logistic regression coefficients by absolute magnitude. `course_portuguese` and `higher_yes` are the strongest positive predictors of passing, while `course_math` and `higher_no` are among the strongest negative predictors. These relationships suggest that certain contextual factors may influence student success more strongly than others.

Another observation was that different training strategies can significantly change how the model behaves. The standard logistic regression model produced the highest F1 score, but the out-of-fold and threshold adjusted models were better at detecting failures. This demonstrates a trade off between predicting passing students accurately and identifying failing students early.

These results also highlight how evaluation metrics can influence how we interpret model performance. A model that achieves the best F1 score overall might not necessarily be the best choice if the main goal is identifying struggling students.

8 Planned Experiments

For future work we plan to evaluate additional machine learning models including neural network approaches. These models may capture nonlinear relationships between features that simpler models cannot represent.

We also plan to explore techniques for addressing the class imbalance present in the dataset. Methods such as class weighting, oversampling of failing students, or under sampling of passing students may improve the model's ability to detect at risk students.

Another direction for future work is performing feature importance analysis. Understanding which student attributes contribute most strongly to predictions could provide useful insights into the factors that influence academic success.

We are also interested in exploring how models behave when different subsets of features are used. For example, it may be useful to evaluate models that only use academic related features compared to models that also include demographic attributes.

9 Team Members and Responsibilities

- Jacob Bettinger – Model implementation and experiments
- Tyler Wolstone – Data preprocessing and evaluation
- Saitej Bommidi – Literature review and writing
- Jason Manhart – Experimental analysis and visualization
- Joseph Miller – Presentation and report coordination

References

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